Region-Based Successor Measures for Offline Robot Manipulation

Anonymous authors
Paper under double-blind review

Abstract

Successor measures enable zero-shot reinforcement learning by predicting where a policy will spend time under any test-time reward. Yet forward–backward (FB) representations perform poorly on offline robot manipulation: on OGBench cube manipulation, FB reaches at most 0.53 success, compared with 0.93 for an architecture-matched goal-conditioned IQL baseline. We identify a mismatch between the FB objective and manipulation data. FB estimates a pointwise occupancy density, but under deterministic dynamics its one-step transition is a point mass and therefore has no density with respect to a non-atomic reference distribution. Manipulation play amplifies this problem because objects remain stationary for long segments, repeatedly producing the same matched transitions. We instead learn discounted occupancy of finite goal regions. This replaces the singular density target with a bounded indicator-reward Bellman target. We first learn an absorbing reachability value for control, then fine-tune a non-absorbing region-occupancy model for reward composition. Region-based TD outperforms the matched GCIQL baseline on all three evaluated manipulation environments and achieves 1.9× FB's average return on 48 held-out rewards. Direct calibration and non-negativity tests evaluate when the learned function behaves as a successor measure and reveal the remaining errors at fine spatial scales.

1 Introduction

A successor measure predicts the discounted occupancy of a set of future states. Once learned, it converts a new reward function into a value function without further environment interaction (Dayan, 1993; Blier et al., 2021). Forward–backward (FB) representations implement this idea by learning a pointwise occupancy density, and have performed well on locomotion and navigation (Touati & Ollivier, 2021; Touati et al., 2023).

The same method performs poorly on offline robot manipulation. On OGBench cube manipulation (Park et al., 2025), FB reaches at most 0.53 success, whereas an architecture-matched GCIQL baseline (Kostrikov et al., 2022) reaches 0.93. We show that the failure comes from estimating occupancy at points rather than over regions. In deterministic systems, the one-step transition is a point mass and has no density with respect to a non-atomic reference distribution. Manipulation play makes this failure especially severe: the object remains stationary on 48% of transitions, repeatedly presenting the same matched transition to FB's non-saturating surrogate loss.

The fix follows from the Bellman equation itself. Written over a finite goal region rather than a point, the successor measure's one-step term is a bounded 0/1 indicator, and the objective becomes goal-conditioned TD (Figure 2). Two backups then correspond to the two questions a region supports: an absorbing backup learns whether the region will be reached, and a non-absorbing backup learns how long it will be occupied. We train the reachability value first, because control depends on reaching, and fine-tune the occupancy model from that checkpoint, because reward composition requires occupancy. The reward readout, including the inverse-region-mass weight it requires, follows from the same change of estimand (§2.5).

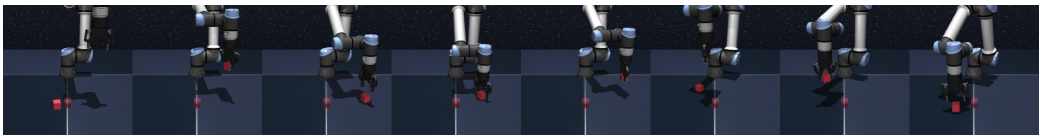


Figure 1: The data behind the failure: one trajectory from cube-single-play-v0, rendered by replaying the dataset's stored simulator states. The collector moves the cube, pauses, regrasps, and returns to the same configurations repeatedly; between manipulations the cube does not move at all. The cube is stationary on 48% of all transitions, so the discounted occupancy a successor measure must predict concentrates at these resting configurations (§3).

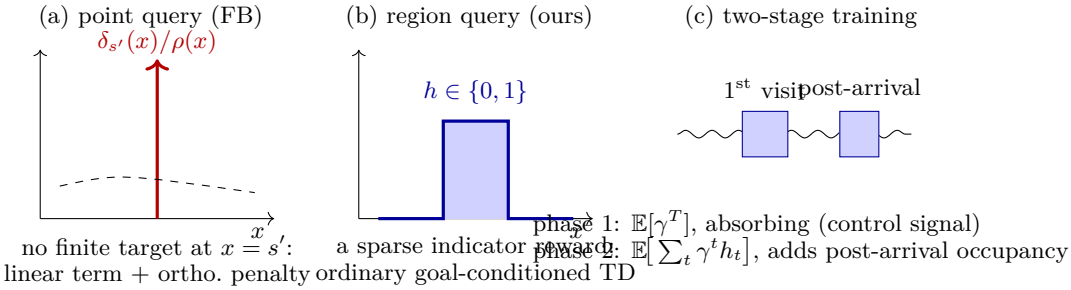


Figure 2: (a) FB regresses a successor density whose empirical target at the observed successor is a Dirac measure — no finite prediction is correct there, so the loss substitutes an unbounded linear term restrained only by an orthonormality penalty; play data supplies those matched pairs on half of all transitions. (b) Querying a region turns the immediate term into a sparse indicator reward and the objective into standard goal-conditioned TD. (c) The two-stage training matches the data's two components: train the absorbing first-visit value first (all control signal), then add the post-arrival occupancy by switching to the non-absorbing backup with the goal representation frozen.

Region-based TD outperforms the matched GCIQL baseline on all three evaluated environments, and the fine-tuned occupancy model achieves 1.9× FB's average return on 48 held-out test-time rewards. Direct calibration and non-negativity tests evaluate when the learned function behaves as a measure. The main input we do not learn is the region itself — a three-dimensional object position and a 4 cm radius, taken from the benchmark; an end-effector-based region retains much of the benefit, and discovering such regions from data is open. Throughout, zero-shot means no optimization on the test reward, given the supplied abstraction and reward labels on offline dataset states.

2 Successor densities versus region occupancy

2.1 Successor measures and forward–backward representations

Goal-conditioned RL. We consider a Markov decision process with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel P, and discount $\gamma \in (0, 1)$. In goal-conditioned RL, a goal g defines the sparse reward $r_g(s') = \mathbf{1}[s' \in S_g]$ for some goal set S_g, and the standard reachability value of a policy π (the goal-conditioned value under the absorbing convention) is $\mathbb{E}_\pi[\gamma^{T_g}]$, where T_g is the hitting time of S_g (Kaelbling, 1993; Andrychowicz et al., 2017). Offline, the agent learns from a fixed dataset of transitions; hindsight relabeling supplies goals from observed states.

Successor measures. The successor measure of π is its discounted state occupancy, viewed as a measure over sets $X \subseteq \mathcal{S}$:

$$M^{\pi}(s, a, X) = \mathbb{E}_{\pi}\Big[\sum_{t\geq 0} \gamma^t \mathbf{1}[s_{t+1} \in X] \;\Big|\; s_0 = s,\, a_0 = a\Big]. \tag{1}$$

Intuitively, $M^{\pi}(s, a, X)$ counts the discounted time the policy will spend in X. Its usefulness is that any reward's value is an integral against it, $Q_r^{\pi}(s, a) = \int r(x)\, dM^{\pi}(s, a, x)$: dynamics are learned once, rewards are supplied later (Dayan, 1993; Blier et al., 2021).

Its Bellman equation, written over sets, is

$$M^{\pi_z}(s, a, X) = P(s' \in X \mid s, a) + \gamma\, \mathbb{E}_{s', a' \sim \pi_z}\big[M^{\pi_z}(s', a', X)\big]. \tag{2}$$

Forward–backward representations and zero-shot RL. FB methods (Touati & Ollivier, 2021; Touati et al., 2023) factorize the successor measure's density with respect to a reference distribution ρ (in practice, the dataset marginal):

$$M^{\pi_z}(s, a, X) \approx \int_X F(s, a, z)^{\top} \psi(x)\, d\rho(x). \tag{3}$$

Here z is a task vector indexing a family of policies π_z, and x is a query state, independent of z. At test time, a reward r observed on dataset samples is converted to a task vector by least-squares reward inference, $z_r \propto \mathbb{E}_{x\sim\rho}[r(x)\psi(x)]$, and the agent acts greedily on $Q_r(s, a) \approx F(s, a, z_r)^{\top} z_r$ with no further training. This is the zero-shot RL protocol we evaluate throughout.

2.2 Why pointwise density estimation is problematic

Proposition 1 (Point densities are singular; region targets are bounded). Suppose the dynamics are deterministic, $s' = f(s, a)$, and the reference distribution ρ is non-atomic. Then the one-step transition measure $\delta_{f(s,a)}$ has no density with respect to ρ. For any measurable region G, however, $P(G \mid s, a) = \mathbf{1}[f(s, a) \in G] \in \{0, 1\}$, and the region-occupancy Bellman operator obtained from Equation 2 is a γ-contraction on bounded functions. (Proof in Appendix A.9.)

The first half is the problem: FB regresses the density $m_z = dM^{\pi_z}/d\rho$, whose one-step Bellman target contains the density of $\delta_{f(s,a)}$ with respect to ρ — which does not exist. The ρ-weighted squared error remains numerically defined, but it is minimized by densities that grow without bound at the observed transitions. Non-atomic means no single state carries positive mass under ρ ($\rho(\{x\}) = 0$ for all x), which holds for any distribution with a density over a continuous state space; in a finite MDP ρ is atomic and the density exists, which is why the failure does not arise in tabular successor representations. The singularity follows from deterministic dynamics alone; it is not created by manipulation data. Manipulation play amplifies its practical effect (§3): long stationary segments repeatedly present the same matched transition to the non-saturating term of FB's surrogate loss.

Standard FB works around the singular term with a surrogate loss (Touati & Ollivier, 2021),

$$\mathcal{L}_{\mathrm{FB}} = \tfrac{1}{2}\mathbb{E}_{x\sim\rho}\big[(m_z - \gamma \bar{m}_z)^2\big] - \mathbb{E}\big[m_z(s, a, s') - \gamma \bar{m}_z(s, a, s')\big] + \lambda_{\mathrm{ortho}}\mathcal{L}_{\mathrm{ortho}}. \tag{4}$$

This has the structure of a contrastive objective: random query states are negatives and receive a squared residual, while the observed successor is the positive pair and receives a term that rewards increasing its score (Eysenbach et al., 2022). But where a contrastive loss saturates its positive term (through a log-sigmoid), the term here is linear: increasing the matched score always reduces the loss, at any scale — and on play data the same matched pairs recur for dozens of consecutive steps. The only component of the objective that opposes the growth is the orthonormality penalty on ψ, a constraint on representation geometry acting as an implicit normalizer.

2.3 Learning occupancy of finite regions

The second half of Proposition 1 is the method. The environment provides a low-dimensional outcome map $c(x)$ (here the object position) and a radius ϵ, which define regions $B_\epsilon(g) = \{x : \|c(x) - c(g)\| \le \epsilon\}$. Define the indicator reward

$$h(s', g) = \mathbf{1}[\|c(s') - c(g)\| \le \epsilon] \in \{0, 1\}, \tag{5}$$

and regress the region occupancy $M(s, a, B_\epsilon(g))$ instead of the density $m(s, a, g)$. Evaluated on the region $B_\epsilon(g)$, the immediate term of Equation 2 is $P(s' \in B_\epsilon(g) \mid s, a)$, whose sample estimate is the indicator $h(s', g)$ — a bounded, sparse, goal-conditioned reward of the kind offline GCRL trains on routinely. The loss is full-matrix TD regression over the batch,

$$\mathcal{L} = \tfrac{1}{n^2} \sum\nolimits_{ij} \big(Q_{ij} - y_{ij}\big)^2, \qquad Q_{ij} = F(s_i, a_i, z_j)^\top \psi(g_j), \tag{6}$$

with no linear term and no orthonormality penalty: the 0/1 reward supplies the scale that the penalty previously had to impose. We have replaced pointwise density regression with a bounded region-occupancy TD objective.

Two backups, one per component. A finite event admits two TD targets, distinguished by what happens after the first hit:

$$y_{ij}^{\text{reach}} = h_{ij} + \gamma(1 - h_{ij})\, \bar{Q}_{ij}, \qquad y_{ij}^{\text{visit}} = h_{ij} + \gamma\, \bar{M}_{ij}. \tag{7}$$

The first treats the region as absorbing, exactly as in standard goal-conditioned RL; the second does not, so occupancy keeps accumulating.

For a fixed continuation policy, both operators are γ-contractions with unique fixed points $Q^{\text{reach}}(s, a, g) = \mathbb{E}[\gamma^{T_g} \mathbf{1}[T_g < \infty]]$ and $M^{\text{visit}}(s, a, g) = \mathbb{E}[\sum_{t \ge 0} \gamma^t h(s_{t+1}, g)]$, where $T_g = \min\{t \ge 0 : s_{t+1} \in B_\epsilon(g)\}$ is the hitting time of $B_\epsilon(g)$ along the successor sequence (Appendix A.9). These are the two components that §3 separates empirically: $Q^{\text{reach}} \in [0, 1]$ is the standard reachability value — the small component that carries the control signal — and $M^{\text{visit}} \in [0, 1/(1-\gamma)]$ is the successor measure of the region, whose magnitude on play data comes mostly from occupancy after the first arrival. We refer to them below as the reachability value and the region occupancy.

2.4 Reachability pretraining and occupancy fine-tuning

The data statistics motivate training them in that order. Phase 1 trains the reachability value with the policy's goal equal to the goal in the TD loss: a standard sparse-reward goal-conditioned RL problem in which every unit of value reflects reaching. Phase 2 warm-starts from that checkpoint, freezes ψ, switches to the non-absorbing backup, and samples the task z and the goal g independently, $M(s, a, z, g) = F(s, a, z)^\top \psi(g)$ — the separated form that reward integration requires (Equation 3). Phase 2 adds the post-arrival occupancy that reward integration needs on top of a goal representation that phase 1 has already validated for control. The two-stage procedure is heuristic: occupancy does not factor exactly into reachability and recurrence for finite regions (Appendix A.7), so phase 2 optimizes occupancy directly with TD rather than composing estimated factors.

Freezing the goal embedding. Phase 2 trains the actor DDPG-style against the occupancy critic while the goal embedding ψ receives no gradient. The freeze preserves the goal representation validated in phase 1: §3.1 shows that training ψ against concentrated targets corrupts it, and collapse under the density objective is twice as fast with ψ trainable as with it frozen. This resembles fixed-feature successor learning (Barreto et al., 2017) and slow-target stabilization in actor–critic methods (Lillicrap et al., 2016), although it is distinct from the ordinary stop-gradient applied to TD bootstrap targets; the connections are discussed in Appendix A.8.

2.5 Test-time reward inference

The readout follows from the same change of estimand. Any reward's value is $Q_r^\pi(s, a) = \mathbb{E}_{g \sim \rho}[r(g)\, m^\pi(s, a, g)]$. For regions small enough that the density is approximately constant

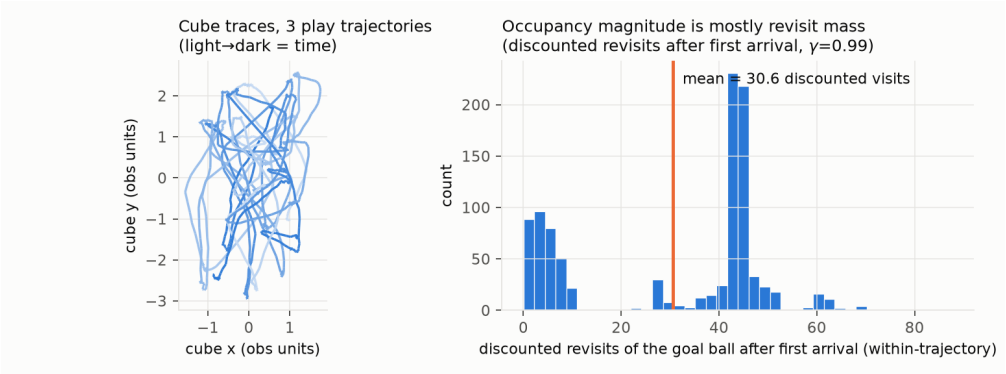


Figure 3: Manipulation play data concentrates occupancy where the object rests. Left: cube traces of three play trajectories — the cube rests, is carried, and rests again. Right: discounted occupancy of a 4 cm goal ball accumulated after its first arrival (within-trajectory Monte Carlo, $\gamma = 0.99$): the mean is 30.6 discounted visits, out of a maximum possible $1/(1-\gamma) = 100$. Most of the magnitude of the successor measure on this data accrues after the first arrival, not from reaching.

on the ball, $M^\pi(s, a, B_\epsilon(g)) \approx m^\pi(s, a, g)\, N_\epsilon(g)$ with $N_\epsilon(g) = \rho(B_\epsilon(g))$, so

$$Q_r^\pi(s, a) \approx \mathbb{E}_{g\sim\rho}\left[\frac{r(g)}{N_\epsilon(g)}\, M^\pi(s, a, B_\epsilon(g))\right], \qquad z_r = \mathbb{E}_{g\sim\rho}\left[\frac{r(g)}{N_\epsilon(g)}\, \psi(g)\right], \tag{8}$$

where the second expression is the induced task embedding for the bilinear model and N_ϵ is estimated by a small trained head $\hat{N}$ (§4.3). Equation 8 has one training-side consequence: inferred embeddings z_r are averages of goal embeddings, so the distribution of z seen in training must include such averages; we add reward-induced embeddings to the task-sampling distribution (§4.2, Appendix A.6).

3 Why the issue matters on manipulation play

What is different about manipulation play data? In the domains where FB was developed — unsupervised-exploration locomotion and maze navigation — the variable that defines tasks is the agent's own configuration, and it changes at essentially every step: a walker's joints never stop, a maze agent keeps moving. Occupancy under such data is diffuse. Manipulation play data is the opposite. The goal-relevant state variable is the object's position, and the object only moves while grasped (Figure 1).

Three statistics quantify this (Figure 3). First, on cube-single-play-v0 the cube position is bit-identical across 48% of all transitions, with a median stationary run of 44 consecutive steps. Second, a 4 cm ball around a random dataset state contains only 1.3% of the data, yet the discounted occupancy a trajectory accumulates in that ball after first arriving averages 25–31 visits out of a possible 100: most of the successor measure's magnitude on this data comes from the object sitting inside the goal region, not from reaching it. Third, post-arrival occupancy is predictable from the goal alone: a regressor given only the goal predicts Monte-Carlo occupancy at rank correlation 0.60–0.63, while the reachability value alone predicts it at 0.06. This suggests that the component dominating the measure's magnitude contains little action-selection information, and that most of that information is in the smaller first-arrival component.

The structure strengthens with every additional object, because play manipulates one object at a time while every other object rests. On cube-double-play-v0, each cube is stationary on 72% of transitions and post-arrival occupancy of the goal region averages 32; on scene-play-v0, the drawer and window are stationary on 91% of transitions, the buttons on 99.6%, and post-arrival occupancy averages 37 — while the conjunctive goal region (all objects in place) covers only 0.03–0.4% of the data (Appendix A.1). If this mechanism is what breaks

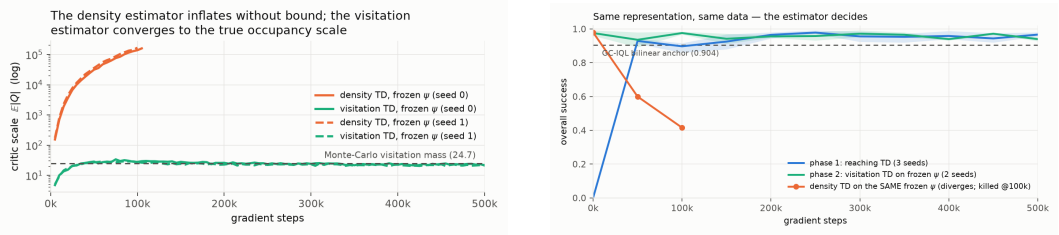


Figure 4: One frozen representation, three objectives. Left: critic scale $\mathbb{E}|Q|$ (log axis) while training on a frozen goal embedding ψ. The density objective (orange) grows without bound once the orthonormality term is removed: its linear term always rewards a larger matched score, and play data supplies the same matched pairs over and over. The indicator-reward objective of §2 (aqua) converges to the Monte-Carlo estimate of the true discounted occupancy (dashed), because every entry of its TD target is a finite number. Right: task success over training under the same frozen embedding: the goal-reaching TD (blue) and the occupancy TD on the frozen phase-1 embedding (aqua) hold the goal-conditioned IQL level for the full run; the density TD (orange) degrades immediately.

density estimation, methods built on occupancy densities should degrade most where the concentration is strongest — and they do: FB scores 0.42 on cube-single but 0.05 on scene (Table 1).

3.1 A frozen-representation experiment isolates the objective

To rule out representation quality as an explanation, we freeze ψ at a checkpoint that already supports 0.97 task success and train only the density estimator on top of it. Without the orthonormality term, the critic's scale grows from 154 to 5.4×10^4 within 65k steps and mean success decreases monotonically from 0.98 to 0.42 by 100k steps, three seeds (Figure 4); letting ψ train merely accelerates the collapse (0.97 $\rightarrow$ 0.07 by 50k). With the orthonormality term, scale is stable but success reaches only 0.225; a fully tuned FB implementation reaches ≈ 0.42, against 0.928 for the architecture-matched GCIQL baseline. The estimator, not the representation, is the bottleneck.

4 Experiments

We organize the evaluation around four questions: does region-based TD restore goal-conditioned control (§4.1); does occupancy fine-tuning improve held-out reward transfer (§4.2); does the learned function satisfy measure properties (§4.3); and does the approach extend to multi-object manipulation (§4.4).

Setting. We use OGBench state-based manipulation (Park et al., 2025), primarily cube-single-play-v0: a 6-DoF arm, one cube, 28-dimensional states, and 10^6 offline transitions of scripted play that repeatedly picks the cube up, carries it, and sets it down. No benchmark reward labels are used to fit any of our objectives. The method is given a state abstraction $c(x) \in \mathbb{R}^3$ (cube position) and a radius $\epsilon = 4\,\text{cm}$, which define goal regions $B_\epsilon(g) = \{x : \|c(x) - c(g)\| \leq \epsilon\}$. This is a substantive assumption — three of twenty-eight state dimensions, at the benchmark's own tolerance — and we quantify what it contributes in §4.1.

Protocol. Unless stated otherwise: rank-50 bilinear critic $F^\top \psi$, twin heads with a minimum target, $\gamma = 0.99$, hindsight-relabeled batch goals (a batch of n transitions yields an $n \times n$ matrix of state–goal pairs), a flow-matching behavior-cloning actor with one-step distillation, 500k gradient steps, 2–3 seeds per cell. Task success means the benchmark's goal-conditioned evaluation, averaged over its five tasks with 50 episodes each. Measure-side evaluations (Monte-Carlo occupancy comparisons, reward-transfer protocols) are defined where they are used.

Method	cube-s	cube-d	scene
reported by OGBench (Park et al., 2025) — reference implementations, 8 seeds			
GCBC	0.06	0.01	0.05
GCIVL	0.53	0.36	0.42
GCIQL	0.68	0.40	0.51
CRL	0.19	0.10	0.19
HIQL	0.15	0.06	0.38
architecture-matched comparison — same data, flow-BC actor, bilinear rank-50 critic, 2–10 seeds			
GCIQL baseline	0.928	0.265	0.281
CRL	0.311	—	—
Tuned FB	0.42	—	0.047
FB (reference reimpl.)	0.532	—	—
FB loss on regions	0.284	—	—
Reachability value (phase 1)	**0.967**	**0.549**	**0.484**
Region occupancy (phase 2)	0.940	0.358	0.342

Table 1: Goal-reaching success (500k steps; five tasks, 50 episodes each). The degenerate puzzle-3x3, where the matched implementation solves only task 1, is omitted. The top block quotes the OGBench paper's reference implementations (their architectures and tuning); the middle block re-implements baselines in the matched architecture, except FB (reference reimpl.), a reimplementation of the reference FB code (single seed). Dashes: not run.

4.1 Does region-based TD restore goal-conditioned control?

Yes. Table 1 compares goal-reaching success on the three environments, against the OGBench reference implementations and against baselines re-implemented in the matched architecture.

Region-based TD restores goal-conditioned control. The reachability value outperforms the architecture-matched GCIQL baseline on all three environments, with the largest improvements on cube-double and scene (Table 1). Against OGBench's reference implementations it is the strongest evaluated method on cube-single and cube-double and comparable to reference GCIQL on scene. With a bounded target, the same representation class that FB uses is competitive with the strongest evaluated baselines.

Successor-style baselines degrade where occupancy concentrates. CRL, the contrastive counterpart, scores 0.10–0.31 across implementations, and FB drops to 0.047 on scene, the environment where the object-at-rest concentration is strongest (§3). This pattern is consistent with the proposed mechanism, although it does not isolate it.

Which change is responsible? A 2×2 ablation crosses the loss form (FB's surrogate vs. indicator-TD) with the query support (point vs. region). FB's own loss on regions reaches 0.284 (Table 1, "FB loss on regions"); indicator-TD with the region collapsed back to a point ($h = I$ on the batch) reaches 0.108; the combination reaches 0.922 (replicated in a second implementation). Neither change suffices alone. The region also need not be the evaluation variable: the same ball built on the end-effector position — correlated with, but distinct from, the evaluated cube — reaches 0.823. The region variable therefore need not exactly match the evaluation variable, although this ablation does not establish that the choice of variable is unimportant in general.

4.2 Does occupancy fine-tuning improve held-out reward transfer?

Yes: fine-tuning nearly doubles compositional-reward return, and with the two corrections below the final model reaches 1.9× the mean return of the reimplemented FB baseline (Table 2).

The rewards are functions of the supplied cube abstraction, all with values in $[0, 1]$: indicator balls at radii 0.4/0.8/1.6 (the training kernel is 0.4), Gaussians at bandwidths 0.4/0.8, and

Method	concentrated	broad	compositional	mean
FB (reference reimpl.)	8.4	35.4	6.9	17.2
FB loss on regions	6.4	27.2	5.0	13.1
Reachability value (phase 1)	12.1	27.8	8.5	16.8
+ occupancy (phase 2)	16.3	28.5	15.8	20.3
+ reward-embedding coverage	20.2	31.5	23.0	24.5
+ measure regularizers	**26.1**	**41.6**	**33.0**	**32.4**

Table 2: Held-out reward return on cube-single: 48 rewards over the cube abstraction, absolute discounted return (rewards in $[0, 1]$, returns in $[0, 86.6]$, behavior-policy baseline 8.1; family means of two seeds, FB single-seed). Concentrated = balls at radii 0.4/0.8 and Gaussians at bandwidth 0.4; broad = ball 1.6 and Gaussian 0.8; compositional = two disjoint zones. Rows below the rule are cumulative; goal-reaching success is unchanged by the two corrections (0.940 → 0.947–0.950).

a two-zone reward (two disjoint balls more than 1.6 apart, not expressible as any single goal); eight instances per family, centers drawn from dataset states. Task vectors come from least-squares reward inference over 10^4 relabeled dataset rows — no ridge term, no candidate search. For per-family analysis we score control as the fraction of the gap between a random-z floor and a single-goal oracle proxy; scores above 1 mean beating that proxy, not optimality.

FB's return comes from broad rewards. The reference-reimplementation row shows FB performing best on the broad family — above our phase-1 and phase-2 models, because a 16 cm ball assigns reward to almost any cube displacement — and 3–5× below the final method on the concentrated and compositional families, where the reward has structure.

Occupancy fine-tuning trades goal-reaching success for transfer. Fine-tuning reduces goal-reaching success on cube-double and scene (−0.19/−0.14; §4.4 removes this reduction) but improves reward transfer where occupancy matters: on the compositional family, which no single goal expresses, return nearly doubles. The two corrections below raise mean return further without reducing goal-reaching success.

Verifying the readout weight. The inverse-mass weight of Equation 8 is visible in the data. With the weight omitted, predicted returns rank well in every family (Spearman 0.62–0.90) but are uniformly 0.014–0.030 of the Monte-Carlo magnitude — a single family-independent scale factor. The mean region mass, measured independently, is $\bar{N} = 0.0137$; the uncorrected predictions lie on $y = \bar{N}x$ (Figure 5), and weighting the readout by $1/\hat{N}(g)$ restores scale to 0.79–2.2 (derivation in Appendix A.9). The correction substantially reduces the scale error, although predictions can still differ from Monte Carlo by up to a factor of two.

Covering reward-induced embeddings. Correcting the scale does not by itself improve behavior — two-zone control falls from 0.86 to 0.46 under the corrected embeddings. This is the training-side consequence anticipated in §2.5: an inferred embedding for a broad reward is an average of many goal embeddings and lies in a part of task space that training, which samples z from random directions and single-goal embeddings, never visits. The test is to add such embeddings to training: a third task-sampling component draws a random synthetic reward and embeds it as inference would. A kernel generator matches the evaluation families' functional forms; an rff generator shares no functional form with any evaluation family and controls for reward-family leakage (Appendix A.6).

Training on reward-induced task embeddings improves control across most reward families (Table 3). The broad indicator ball, at 0.07 at baseline, recovers to 0.49–0.55 in every arm — including rff, whose training rewards share no form with the evaluation family: the improvement comes from populating the averaged region of task space, not from familiarity with the test rewards. The compositional family reaches 1.35, exceeding the better single-zone proxy by 35%, which is what reward composition should produce. Combined with the

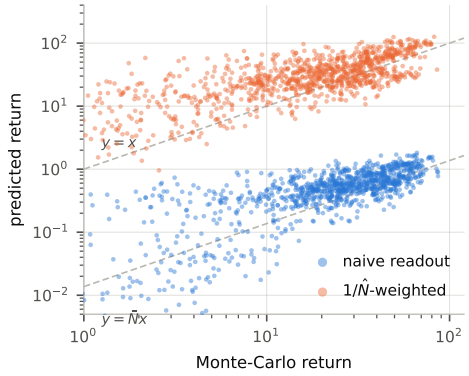


Figure 5: Predicted vs. Monte-Carlo return on all 48 held-out rewards, log–log. Uncorrected reward inference tracks $y = \bar{N}x$ with $\bar{N} = 0.0137$ the independently measured mean region mass; the $1/\hat{N}$-weighted readout tracks $y = x$.

Family	baseline	kernel η=0.25	kernel η=0.5	rff η=0.25
ball 0.4	0.80	0.87	0.87	0.79
ball 0.8	0.66	0.71	**0.95**	0.73
twozone	0.86	0.92	**1.35**	1.11
smooth 0.4	0.50	0.49	**0.78**	0.55
smooth 0.8	0.15	0.27	0.33	0.12
ball 1.6	0.07	**0.53**	**0.49**	**0.55**
mean (all 6)	0.507	0.630	**0.795**	0.640
task success	0.887	0.948	0.950	0.930
set-mass calibration check (hw 0.8)	1.10/1.16	1.10/1.19	1.23/1.31	1.26/1.12

Table 3: Zero-shot control (normalized return relative to random-z and single-goal-oracle baselines) after adding reward-derived task embeddings to training at mixing fraction η; two-seed means. The bottom rows check that the intervention reduces neither task success nor calibration. Oracle ceilings agree within 3–23% across these runs, so normalized scores are comparable among them.

regularizers of §4.3, the broad Gaussian improves from 0.24/0.43 to 0.46–0.86 across seeds, though it remains the least reliable family. Task success improves alongside (0.887 → 0.950) and calibration is intact. Figure 6 resolves the same result by the reward's spatial support.

What occupancy fine-tuning adds over the reachability value. Table 2 answers on returns; the mechanism is visible when both policies are evaluated with the goal supplied directly: the phase-2 policy accumulates 1.13–2.00× more reward than the phase-1 policy on identical rewards. A reachability value is saturated once the region is entered, so its greedy policy enters and stops; a policy trained on occupancy returns to and remains in the rewarded region — exactly the post-arrival occupancy that §3 showed dominates this data. The advantage disappears on the broadest rewards (0.99–1.05×), where remaining in place earns nothing extra. (The phase-2 checkpoint continues training from the phase-1 one, so the comparison is not iso-budget; the goal-supplied comparison is less exposed to this, but an equal-training-budget control has not been run.)

4.3 Does the learned function satisfy measure properties?

Only partially: the estimator is coarsely calibrated but carries substantial negative mass, and removing the negative mass requires both regularizers of this section (Table 5). Strong task success could not have told us this: control depends mainly on the critic's action ranking at the current state. Empirically, the goal-reaching and occupancy critics select

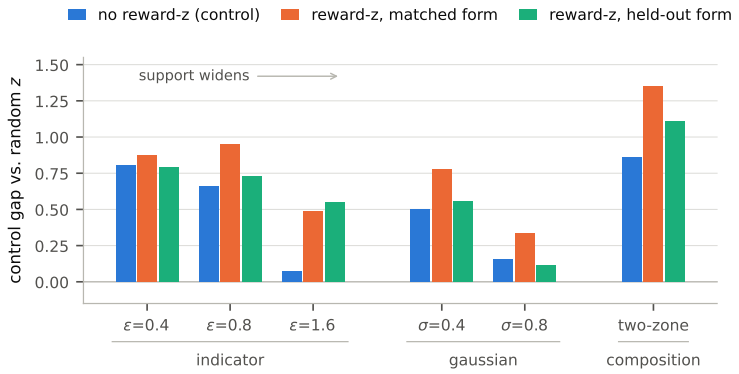


Figure 6: Zero-shot control against the reward's spatial support (normalized return as in Table 3). Control degrades as support widens within each family. Adding reward-induced embeddings to the task-sampling distribution recovers the wide indicator ball and the compositional family, but not the wide Gaussian; the held-out reward form (rff) performs comparably to the matched one on the recovered families.

box half-width	model/MC (exact N)	model/MC (trained $\hat{N}$)	Spearman
0.4	3.22/2.03	1.94/1.22	$+0.31/+0.33$
0.8	1.92/2.08	**1.10/1.16**	$+0.68/+0.72$
1.6	2.33/2.18	**1.23/1.14**	$+0.85/+0.82$

Table 4: Calibration per seed: predicted over Monte-Carlo set mass (ratio; 1.0 is perfect) and rank correlation across boxes. Monte Carlo truncates at H=200, omitting $\gamma^H = 0.134$ of the discounted tail, so ratios near 1.15 are near-exact. Coarse boxes are calibrated; the smallest scale is not.

the same greedy action on only 29.5%/29.9% of a grid of 24 states × 150 goals × 16 candidate actions (chance 6.25%). Task success provides evidence that the reachability critic induces an effective policy; it does not establish calibration of M^{visit}. We therefore test the occupancy estimator directly for the properties a measure must have: calibrated set mass and non-negativity.

4.3.1 Set-mass calibration

We compare predicted occupancy of random boxes against Monte-Carlo rollouts of the same policy (32 starts × 3 rollouts, horizon 200). The learned quantity is a region mass, so converting it to a density requires dividing by the region's probability under the data distribution, $N(g) = \rho(B_\epsilon(g))$; a small trained head $\hat{N}(g)$ estimates this factor.

At half-widths 0.8 and 1.6 the corrected estimate is within 10–23% of Monte Carlo with rank correlation 0.68–0.85 (Table 4). Two qualifications: with exact counts in the denominator, the uncorrected estimate is 1.9–3.2× too large (the head's +28% bias partially compensates), and the smallest scale — the training kernel's own — remains 1.2–1.9× off on every variant we trained, consistent with a resolution limit of the rank-50 factorization (§6).

4.3.2 Non-negativity and mass conservation

A measure is non-negative. The learned occupancy estimator is not: across 32 policies and 20k dataset goals, about half of all values are negative, the negative values carry 24–29% of the total absolute mass at roughly a third of the positive values' magnitude, and the violation is largest in the densest parts of goal space (0.55–0.59 of values negative in the top density quartile vs. 0.45 in the bottom). These are not boundary fluctuations; they are

Model	negative-mass share	set-mass ratio (hw 0.8/1.6)	total mass
unregularized	0.24–0.29	1.10–1.23	unconstrained
non-negativity only	< 0.01	—	2.5–3.6×
non-negativity + mass	0.010–0.014	1.15–1.29	1.002

Table 5: Measure diagnostics on cube-single (two seeds). Negative mass: share of total absolute mass carried by negative values. Set-mass ratio: predicted over Monte-Carlo region mass with the trained $\hat{N}$ (Table 4 resolves it by scale). Total mass: training-time total-occupancy ratio (target 1). The non-negativity penalty alone removes sign errors but inflates total mass; adding the mass-conservation penalty fixes both.

systematic oscillations where the regression is most constrained. Calibration and invalidity coexist because signs cancel inside a box.

Replacing the discontinuous 0/1 reward with a smooth kernel of matched mass changes nothing (negative fraction 0.50–0.52; Appendix A.5), so the violation is slack in the regression, not an artifact of the target's sharpness.

The direct correction is a one-sided penalty $\lambda\,\mathbb{E}[\mathrm{relu}(-M)^2]$. Alone, it fails, and the failure identifies the missing constraint: at $\lambda \in \{1, 10\}$ the negative fraction drops to 3.1%/0.7%, but total mass inflates to 2.5–3.6× Monte Carlo. The penalty raises negative values, and no term in the objective constrains the total. The missing constraint is a property of the region occupancy's fixed point: because the cube is always somewhere, averaging region mass over goals, each weighted by the inverse of its region's data probability, counts total discounted time,

$$\mathbb{E}_{g\sim\rho}\left[\frac{M(s,a,g)}{N(g)}\right] = \sum_{t\geq 0}\gamma^t = \frac{1}{1-\gamma} \quad\Longrightarrow\quad \mathcal{L}_{\text{mass}} = \left((1-\gamma)\,\tfrac{1}{n}\textstyle\sum_j M_{ij}/\hat{N}_j - 1\right)^2. \quad (9)$$

The penalty holds the model to this identity (exactness conditions in Appendix A.9). With both regularizers, sign and scale are fixed jointly (Table 5) and task success is unchanged (0.94–0.95 vs. control 0.94–0.96).

The measured cost: rank correlation at the middle calibration scale drops from 0.68 to ≈ 0.50 (the coarse scale is unaffected); penalty-only runs already showed this degradation, implicating the penalty strength; a weaker penalty is under evaluation (§6).

4.4 Does the approach extend to multi-object manipulation?

Yes, with one change to how goals are sampled for the occupancy TD; the same change removes phase 2's reduction in goal-reaching success ($-0.19/-0.14$, Table 1). On cube-double the trained region is a conjunction (both cubes within ϵ), so a goal sampled uniformly from the dataset gives the occupancy TD a nonzero indicator at rate 0.046% — 28× sparser than cube-single. We tested the other candidate causes — kernel width, factorization rank, freezing the actor through the phase switch, and their compositions; eight pre-specified ablations, none of which improved success — and one change did: sampling the occupancy TD's goals by hindsight relabeling (future states of the same trajectory), which makes the indicator nonzero at the state–goal pairs the actor loss uses, by construction. The cost: without an importance correction the hindsight-sampled object is no longer the successor measure with respect to the dataset distribution ρ (§4.3.1), and on cube-single — where dataset-sampled goals already give a nonzero indicator at 1.3% — the change reduces success by 0.17–0.24. It is an environment-conditional modification, not a default.

Table 6 scores zero-shot control on the multi-object environments with the cube-single reward-transfer evaluation protocol extended to conjunctive regions (region centers drawn from visited states; the extended protocol reproduces the cube-single numbers of §4.2 exactly). Protocol as before: least-squares reward inference over relabeled rows, nearest-neighbor decoding of z_r to a dataset goal state, the benchmark's success criteria, 25 episodes per task at 500k steps.

		benchmark tasks ↑		region reward ↑	
Env	checkpoint	zero-shot	direct GC	final	anytime
cube-s	reachability value	0.968	0.967	0.741 (R1–R3 mean)	
cube-d	reachability value	0.480	0.504	0.15	0.23
	+ occupancy, dataset goals	0.312	0.368	0.04	0.04
	+ occupancy, hindsight goals	0.488	0.500	0.08	0.18
scene	reachability value	0.520	0.468	0.26	0.42
	+ occupancy, dataset goals	0.376	0.332	0.21	0.35
	+ occupancy, hindsight goals	**0.872**	0.916	**0.45**	**0.74**

Table 6: Zero-shot control beyond cube-single. "Zero-shot" infers the task from reward-labeled dataset rows; "direct GC" is the benchmark's goal-conditioned evaluation of the same checkpoint. Region rewards are conjunctive balls at the trained radii around visited-state centers, scored at the final step and at any step; the cube-single region row uses the single-object region protocol and is not comparable to the multi-object cells. Single-seed results; the hindsight direct-GC numbers replicate at three seeds (0.531 cube-d, 0.927 scene, Table 1).

5 Related work

Successor representations and measures. Successor representations predict discounted future occupancy (Dayan, 1993); successor measures generalize them to continuous spaces as measures, or densities against a reference distribution (Blier et al., 2021); FB representations factorize that density and infer tasks by least-squares reward regression (Touati & Ollivier, 2021; Touati et al., 2023). We keep FB's bilinear form and zero-shot protocol and change the estimand from a pointwise density, whose empirical Bellman target is a Dirac measure, to region occupancy, whose target is a sparse indicator reward — a change whose necessity is specific to data that concentrates occupancy at resting states (§3).

Goal-conditioned RL. Goal-reaching values with sparse indicator rewards and hindsight relabeling are the standard objects of offline GCRL (Kaelbling, 1993; Andrychowicz et al., 2017; Kostrikov et al., 2022; Park et al., 2023). Contrastive RL (Eysenbach et al., 2022) learns classifiers of future states rather than densities, avoiding the same singular target from a different direction; our diagnosis in §3 makes the connection explicit — FB's surrogate is contrastive in structure but lacks the saturating positive term that keeps classification bounded. A goal-conditioned value evaluates each goal under that goal's own policy; a successor measure retains an independent query and must support reward composition, which is why our evaluation separates reachability (task success) from the measure claim (calibration, sign, transfer).

Hitting values versus occupancy. The absorbing and non-absorbing TD targets answer different questions — whether a region is reached versus how long it is occupied. Manipulation benchmarks score first arrival into a tolerance region, so a reachability value suffices for control; Table 1 shows occupancy is nonetheless the object that improves reward transfer.

6 Limitations and open problems

The region is supplied. c and ϵ come from the benchmark's success criterion. The 2×2 ablation bounds what this buys — the region alone rescues nothing (0.284 under FB's loss), and a correlated non-evaluated region retains most of the benefit (0.823) — but discovering outcome variables and scales from data is unsolved here. All 48 transfer rewards are functions of the supplied abstraction; "zero-shot" refers to held-out rewards, not held-out abstractions.

Multi-object evidence covers success, not calibration. Zero-shot control now has multi-object evidence (§4.4), and phase 2's success reduction on cube-double/scene is repaired by

hindsight goal sampling — but the occupancy-calibration evidence is weaker there: set-mass ratios are 1.57/1.14 with rank correlations 0.37/0.50 and 71–78% of probe centers degenerate, and return calibration holds only at 2–4× the trained radius. The hindsight-sampled object that restores control is not the successor measure with respect to ρ; whether one object can be both a valid measure and a critic for policy improvement on these environments — by mixing query distributions, or by giving the actor its own hindsight-trained critic — is the open question.

Small-scale calibration, and the cost of the regularizers. Set mass at the training kernel's own scale is 1.2–1.9× off on every variant, unchanged by kernel smoothing or penalties — consistent with a resolution limit of the rank-50 factorization; raising the rank is the natural next experiment. The regularizers cost mid-scale rank correlation ($0.68 \rightarrow 0.50$); a weaker penalty is under evaluation.

Comparisons. The OGBench block of Table 1 is quoted from the paper's reference implementations (8 seeds, their architectures and tuning) and is not an apples-to-apples ablation of our stack — notably, reference GCIQL solves puzzle-3x3 (0.95) where our matched bilinear-flow stack does not. None of the baselines has been run through the reward-transfer evaluation (Table 1 dashes), so the return columns compare only our objectives against FB's loss at matched support. The phase-2-vs-phase-1 return comparison does not use matched training budgets; the oracle proxy understates ceilings for broad and multimodal rewards; the random-z floor is a single draw.

Statistical power; scope of theory; one actor class. Most cells have two or three seeds. The contraction result covers fixed-policy operators, not the full system with a changing actor and twin-min targets. All primary cells share one flow-matching BC actor; a single-seed sweep with a different actor reversed one comparison at one setting, so actor isolation remains open.

7 Conclusion

Manipulation play data has a structure that the domains where zero-shot RL methods were developed do not: the manipulated object is at rest half the time, so most of the discounted occupancy accrues while the object rests and carries no control signal. A pointwise density objective has no finite regression target exactly there, and the standard surrogate fails on a frozen, known-good representation with and without its orthonormality term. Matching the algorithm to the data resolves the failure: region queries make the objective ordinary goal-conditioned TD, and a two-stage procedure trains the reachability value, which carries the control signal, before adding the post-arrival occupancy that reward integration needs. Evaluated jointly on goal-reaching and reward transfer (Table 1), the result beats the matched goal-conditioned baseline on every environment while earning 1.9× FB's return on held-out rewards — with two further corrections, an inverse-mass weight for return prediction and reward-derived task vectors for policy coverage, accounting for most of the remaining transfer gap. The remaining supplied input is the region itself — three coordinates and a radius — and learning it from data is the problem to solve next.

References

Marcin Andrychowicz et al. Hindsight experience replay. In Advances in Neural Information Processing Systems (NeurIPS), 2017.

Andre Barreto et al. Successor features for transfer in reinforcement learning. In Advances in Neural Information Processing Systems (NeurIPS), 2017.

L. Blier, C. Tallec, and Y. Ollivier. Learning successor states and goal-dependent values: A mathematical viewpoint. arXiv preprint arXiv:2101.07123, 2021.

Vivek S. Borkar. Stochastic approximation with two time scales. Systems & Control Letters, 29(5):291–294, 1997.

Peter Dayan. Improving generalization for temporal difference learning: The successor representation. Neural Computation, 1993.

Benjamin Eysenbach, Tianjun Zhang, Sergey Levine, and Ruslan Salakhutdinov. Contrastive learning as goal-conditioned reinforcement learning. In Advances in Neural Information Processing Systems (NeurIPS), 2022.

Leslie Pack Kaelbling. Learning to achieve goals. In International Joint Conference on Artificial Intelligence (IJCAI), 1993.

Vijay R. Konda and John N. Tsitsiklis. Actor-critic algorithms. In Advances in Neural Information Processing Systems (NeurIPS), 2000.

Ilya Kostrikov, Ashvin Nair, and Sergey Levine. Offline reinforcement learning with implicit q-learning. In International Conference on Learning Representations (ICLR), 2022.

Timothy P. Lillicrap, Jonathan J. Hunt, Alexander Pritzel, Nicolas Heess, Tom Erez, Yuval Tassa, David Silver, and Daan Wierstra. Continuous control with deep reinforcement learning. In International Conference on Learning Representations (ICLR), 2016.

Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, et al. Human-level control through deep reinforcement learning. Nature, 518(7540):529–533, 2015.

Seohong Park, Dibya Ghosh, Benjamin Eysenbach, and Sergey Levine. HIQL: Offline goal-conditioned RL with latent states as actions. In Advances in Neural Information Processing Systems, 2023.

Seohong Park, Kevin Frans, Benjamin Eysenbach, and Sergey Levine. Ogbench: Benchmarking offline goal-conditioned reinforcement learning. In International Conference on Learning Representations (ICLR), 2025.

Richard S. Sutton and Andrew G. Barto. Reinforcement Learning: An Introduction. MIT Press, 2nd edition, 2018.

Ahmed Touati and Yann Ollivier. Learning one representation to optimize all rewards. In Advances in Neural Information Processing Systems (NeurIPS), 2021.

Ahmed Touati, Jérémy Rapin, and Yann Ollivier. Does zero-shot reinforcement learning exist? In International Conference on Learning Representations (ICLR), 2023.

A Additional experimental detail

A.1 Play-data statistics

Computed on cube-single-play-v0 (10^6 transitions), cube slice of the raw observation. Stationarity: $\|\Delta c\| < 10^{-6}$ on 47.9% of within-episode transitions (51% below 10^{-3}); stationary runs have median length 44 steps (mean 43.9, p90 47). Region mass: a 4 cm ball around a random dataset state contains on average 1.34% of the data ($\bar{N} = 0.0137$). Post-arrival occupancy: discounted occupancy of the 4 cm goal ball accumulated after first arrival has dataset-level mean ≈ 24.7 (within-trajectory estimate ≈ 30.6, bimodal: passed-through vs. settled). A goal-only regressor of this mass fit on the frozen $\psi(g)$ predicts Monte-Carlo occupancy at Spearman 0.60–0.63 uniformly across first-arrival-time buckets; the reachability value alone predicts at 0.06.

Multi-object play datasets (200 episodes sampled per dataset, conjunctive regions from the environments' success criteria). cube-double-play-v0: each cube stationary on 72% of transitions (all objects simultaneously on 48%); conjunctive region mass 3.3×10^{-4}; post-arrival occupancy of the conjunctive region mean 32.2 (median 43). scene-play-v0: cube stationary 74%, drawer 91%, window 91%, buttons 99.6% (all objects 60%); conjunctive region mass 3.7×10^{-3}; post-arrival occupancy mean 37.0 (median 37.6).

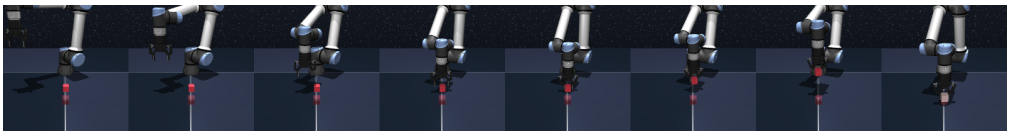


Figure 7: A successful rollout of the fine-tuned occupancy policy (cube-single, benchmark task 2): approach, grasp, transport, and release inside the goal region.

A.2 Divergence: the full control set

Section 3 reports the frozen-representation experiment in condensed form. The complete set of controls:

- Frozen ψ, density surrogate. Critic scale $154 \to 5.4 \times 10^4$ within 65k steps; mean success decreases monotonically from 0.98 (initial) through 0.60 to 0.42 by 100k steps (three seeds). The representation comes from a checkpoint supporting 0.97 control and is held fixed, so representation quality cannot account for the outcome.
- Trainable ψ, density surrogate. Collapses faster: $0.97 \to 0.07$ and $0.98 \to 0.06$ by 50k (two seeds).
- Frozen ψ, indicator-reward TD. Scale converges to the Monte-Carlo estimate of true discounted occupancy; both seeds complete 500k steps without instability.
- Orthonormality retained. Scale bounded; success 0.225.

A.3 Qualitative rollout

A.4 A tabular instance of the same structure

The estimator claims can be exhibited exactly in a 50-state MDP with the same anatomy as manipulation play data: a 7×7 grid plus a pit, where the column is the outcome variable (the region coordinate) and the row is an interaction mode invisible to the region. A 2×2 sticky bowl models a cube at rest — the dwell structure of §3 in miniature; the top row is a conveyor drifting two columns per step regardless of action (a carried cube whose position cannot be held); riding off its end drops into a hard-to-leave pit.

Fitting a rank-8 bilinear $F^\top \psi$ on 20k offline transitions with identical architecture and data, indicator-reward regression reaches relative error 0.05–0.09 and per-state goal-ranking correlation 0.96–0.99; the density-TD form reaches only 0.38 and 0.65, and adding the orthonormality regularizer makes it worse (0.53 and 0.45). With no actor in the loop the density estimator fits poorly but does not diverge, consistent with the divergence of §3 being a property of the estimator-plus-actor system. The environment also exhibits the ranking disagreement that motivates the occupancy objective: with a stable and a transient region weighted to be equally reachable, ranking by reachability value selects the transient region while ranking by occupancy selects the stable one; exact policy evaluation confirms the latter (96.8 vs. 47.9 discounted return).

A.5 Non-negativity and mass-conservation diagnostics, in detail

Negativity measurement: 32 draws of (s_0, a_0, z) with z the projected embedding of a random dataset state and a_0 the actor's own action; 20k dataset goals per draw. Reported: fraction of negative values, share of $\sum |M|$ carried by negative values, both split by quartile of the goal's kernel density $N(g)$ over 10^5 reference rows. Baseline: negative fraction 0.48–0.53, negative mass share 0.24–0.29, rising with density quartile. Penalty only (λ=1/10): negative fraction 3.1%/0.7%, box mass 2.5–3.6× MC. Penalty (λ=1) + mass conservation (coefficient 1): negative fraction 0.078–0.096, negative mass share 0.010–0.014, flat across quartiles; training total-mass ratio 1.002; set-mass ratios (trained-$\hat{N}$) 1.15–1.29 at half-widths 0.8/1.6; Spearman 0.47–0.51 (hw 0.8) and 0.83–0.86 (hw 1.6); task success 0.940/0.953 vs. control 0.937/0.963 under the same parser. Kernel control: a C^1 biweight kernel $(1 - (d/h)^2)^2$,

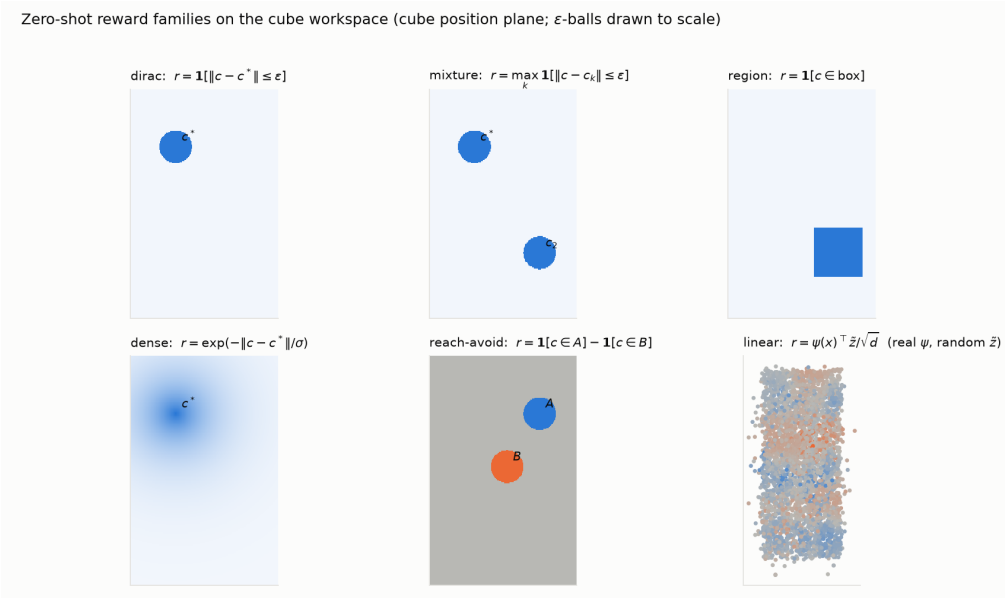


Figure 8: The held-out reward families on the cube workspace (cube-position plane; ϵ-balls to scale). The first five panels are functions of cube position. The linear panel colors dataset states by $\psi(x)^\top \tilde{z}$ for one random direction of the trained embedding: the field is spatially incoherent in cube space, which is the visual form of the coverage problem of §2.5.

$h = 0.6$ (mass-matched to the 0.4 ball), leaves the negative fraction at 0.50–0.52 and does not improve small-box ratios.

A.6 Reward construction

All transfer rewards are functions of the cube position $c(x)$, with centers drawn uniformly from dataset states; eight instances per family; all values in $[0, 1]$. ball ϵ is $\mathbf{1}[\|c(x) - c_0\| \le \epsilon]$ for $\epsilon \in \{0.4, 0.8, 1.6\}$ in observation units (the training kernel is 0.4); smooth σ is $\exp(-\|c(x) - c_0\|^2/2\sigma^2)$ for $\sigma \in \{0.4, 0.8\}$; twozone is $\max_k \mathbf{1}[\|c(x) - c_k\| \le 0.4]$ with $\|c_1 - c_2\| > 1.6$. Inference uses $m = 10{,}000$ relabeled rows. Prediction is scored from 12 teleported starts $\times$ 2 rollouts; control over 10 episodes per drive at horizon 200, so returns lie in $[0, \sum_{t<200} \gamma^t] = [0, 86.6]$. Table 1's column groups average family means: concentrated = {ball 0.4, ball 0.8, smooth 0.4}, broad = {ball 1.6, smooth 0.8}, compositional = twozone. Synthetic training rewards (§4.2) are drawn per batch: kernel mixes balls, Gaussians, and two-zone rewards with centers from the batch and log-uniform radii in $[0.2, 2.0]$; rff draws $r = \text{relu}(\cos(w^\top c + \varphi))$ with $w \sim \mathcal{N}(0, I)/b$, $\varphi \sim \text{U}[0, 2\pi)$. Both are normalized before reward inference.

A.7 The two-stage procedure is not an exact factorization

For a singleton goal, occupancy factors as first-arrival probability times a recurrence term. For finite regions this identity fails: the region can be re-entered from many boundary states, so recurrence depends on where the region was entered and on the actions taken inside it. Empirically, recurrence remains action-dependent on our data: across a grid of states, goals, and candidate actions, the intraclass correlation of recurrence across actions is 0.24–0.29, far from the action-independence an exact factorization would need. The two-stage procedure therefore uses phase 1 only to initialize phase 2, which optimizes occupancy directly with TD.

A.8 Freezing one factor: relation to standard stabilization methods

Holding part of a value function fixed while the rest trains appears in several standard forms. Semi-gradient TD stops gradients through the bootstrap target (Sutton & Barto, 2018). Deep value learning holds a target copy of the network fixed between synchronizations (Mnih et al., 2015), and DDPG trains its actor against a critic whose target copies move on a slower timescale (Lillicrap et al., 2016). Two-timescale analyses of actor–critic make the requirement precise: convergence arguments hold when one component evolves slowly enough to appear stationary to the other (Borkar, 1997; Konda & Tsitsiklis, 2000). Successor-feature methods fix the feature map and train the other factor of a factored value function by TD (Barreto et al., 2017). Phase 2's freeze differs from all of these in one respect: the goal embedding is fixed for the entire phase rather than updated slowly, which is closest to the successor-feature setting.

A.9 Derivations

Proof of Proposition 1. No density: a measure has a density with respect to ρ only if it is absolutely continuous with respect to ρ. The point mass $\delta_{f(s,a)}$ assigns mass 1 to the set $\{f(s,a)\}$, which has ρ-measure zero when ρ is non-atomic, so $\delta_{f(s,a)} \not\ll \rho$ and no Radon–Nikodym derivative exists. Boundedness and contraction: for measurable G and deterministic dynamics, $P(G \mid s,a) = \mathbf{1}[f(s,a) \in G] \in \{0,1\}$. Both region operators act on bounded functions with the sup norm. For the absorbing operator $(\mathcal{T}^{\text{reach}}Q)(s,a,g) = \mathbb{E}[h + \gamma(1-h)Q(s',a',g)]$ and any Q_1, Q_2, $|\mathcal{T}^{\text{reach}}Q_1 - \mathcal{T}^{\text{reach}}Q_2| \le \gamma\,\mathbb{E}[(1-h)\,|Q_1 - Q_2|] \le \gamma\|Q_1 - Q_2\|_\infty$ since $0 \le 1-h \le 1$; the non-absorbing operator is a γ-contraction by the same bound without the $(1-h)$ factor. Banach's fixed-point theorem gives a unique fixed point for each. For the non-absorbing operator, unrolling the recursion and using the Markov property gives $M(s,a,g) = \mathbb{E}[\sum_{t\ge 0}\gamma^t h(s_{t+1},g)]$, bounded by $1/(1-\gamma)$. For the absorbing operator, let $T_g = \min\{t \ge 0 : s_{t+1} \in B_\epsilon(g)\}$ and $Q^* = \mathbb{E}[\gamma^{T_g}\mathbf{1}[T_g < \infty]]$. If $s_1 \in B_\epsilon(g)$ then $T_g = 0$ and the right side of the backup contributes $h = 1 = \gamma^0$; otherwise the backup contributes $\gamma\,\mathbb{E}[\gamma^{T'_g}\mathbf{1}[T'_g < \infty]]$ with T'_g the hitting time from s_1, and $T_g = T'_g + 1$ on that event. So Q^* satisfies the fixed point equation, and by uniqueness it is the fixed point. □

The total-occupancy identity and its exactness conditions. For any start (s,a), exchanging the expectation over goals with the sum over time,

$$\mathbb{E}_{g\sim\rho}\left[\frac{M^\pi(s,a,B_\epsilon(g))}{N(g)}\right] = \mathbb{E}_\pi\Big[\sum_{t\ge 0}\gamma^t\, w(s_{t+1})\Big], \qquad w(x) := \int_{B_\epsilon(x)}\frac{\rho(dg)}{N(g)},$$

using the symmetry $\mathbf{1}[x \in B_\epsilon(g)] = \mathbf{1}[g \in B_\epsilon(x)]$. If goal regions partition the space (disjoint cells), $w \equiv 1$ and the identity $\mathbb{E}_g[M/N] = 1/(1-\gamma)$ is exact. For overlapping balls, $w(x) = 1$ whenever N is constant on $B_\epsilon(x)$, so the identity holds to first order when N varies slowly at scale ϵ; and $\mathbb{E}_{x\sim\rho}[w(x)] = \int \rho(dg)\,N(g)/N(g) = 1$ exactly by Fubini, so the identity is exact on average over states drawn from ρ. The training penalty enforces the batch-mean form, which sits between these regimes.

The $1/N$ readout weight. Any reward's value is $Q_r(s,a) = \mathbb{E}_{x\sim\rho}[r(x)\,m(s,a,x)]$ with $m = dM/d\rho$ wherever the density exists. The trained head outputs region mass, $F^\top\psi(g) \approx M(s,a,B_\epsilon(g)) = \int_{B_\epsilon(g)} m\,d\rho \approx m(s,a,g)\,N(g)$ for m approximately constant on the ball. Hence $m(s,a,g) \approx F^\top\psi(g)/N(g)$ and $Q_r(s,a) \approx \frac{1}{n}\sum_j r(g_j)\,F^\top\psi(g_j)/N(g_j)$ over goals $g_j \sim \rho$. Omitting the $1/N(g_j)$ factor multiplies the estimate by approximately $\bar{N} = \mathbb{E}[N]$ when N is uncorrelated with $r \cdot m$ across goals — the $y = \bar{N}x$ line of Figure 5.

A.10 Monte-Carlo evaluation protocol

Teleport-style Monte-Carlo evaluation of the kind used in §4.3.1 must reset the environment before every rollout, not merely overwrite the simulator state: the episode step counter otherwise persists across rollouts and every rollout after the first truncates immediately,

inflating model-to-Monte-Carlo ratios by one to two orders of magnitude. A useful invariant when debugging such a harness: the implied total Monte-Carlo mass must approach $\sum_k \gamma^k$, because the tracked object is always somewhere.